Simple identities Pt. 1

- Using the definition of expectation in terms of the pdf, $E[f(X)] = \int_{-\infty}^{\infty} f(x)p_X(x)dx$, show that
  - $E[f(W) + g(U)] = E[f(W)] + E[g(U)]$, for any two random variables U and W,
  - $E[X + a] = E[X] + a$, where a is a constant while X is a random variable,
  - $E[aX] = aE[X]$,
  - $var(aX) = a^2 var[X]$, where $var[]$ means variance.
- Use the linearity property of expectation to show that the variance of the sum of two random variables $Z = X + Y$ is given by

$$var(Z) = var(X) + var(Y) + 2C_{XY}$$

(Hint: $var(Z) = E[(Z - E[Z])^2] = E[(X + Y - E[X + Y])^2]$

Pseudo-random numbers

- Homework reading: Find an online source to understand how random numbers are generated on a computer
- Example: https://en.wikipedia.org/wiki/Pseudorandom_number_generator
- Use the `help` for the basic pseudo-random number generators `rand` and `randn` in Matlab
- Write a Matlab script to generate 10,000 trial values from each of these functions and plot their histograms using the `histogram` function
- Use the help for the `rng` function in Matlab to set a seed for the pseudo-random number generator
- Generate 2 sets of 5,000 trial values from `randn` but set the same seed using `rng` before generating each set. Compare the values in the two sets. Do they repeat?

PDF of a Sum

Let X and Y be two random variables, and let the random variable Z be their sum, so that $Z = X + Y$. Then, $F_Z(z)$, the CDF of the variable Z, would give the probabilities associated with that random variable. But by the definition of a CDF, $F_Z(z) = P(Z \leq z)$, and we know that $z = x + y$. Therefore, we need to consider those x and y values whose sum is less than or equal to z, or alternatively, when $y \leq z - x$.

Now X and Y will have a joint PDF, which we shall call $f_{XY}(x, y)$. Probabilities for the joint function are found by integrating the PDF, and we are interested in those probabilities for which $y \leq z - x$. In other words, we have

$$F_Z(z) = P(Z \leq z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{XY}(x, y)\, dy\, dx$$

By using the substitution $y = v - x$, we can then interchange the order of integration.

$$\begin{aligned} F_Z(z) &= \int_{-\infty}^{\infty} \int_{-\infty}^{z} f_{XY}(x, v - x)\, dv\, dx \\ &= \int_{-\infty}^{z} \int_{-\infty}^{\infty} f_{XY}(x, v - x)\, dx\, dv \end{aligned}$$

Taking the derivative of the CDF with respect to z, we obtain the following PDF.

$$f_{X+Y}(z) = \int_{-\infty}^{\infty} f_{XY}(x, z - x)\, dx$$

When the random variables X and Y are independent, then the joint PDF may be written as a product of the PDFs, and we obtain the following result.

If X and Y are independent,
then
$$f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x)\, dx$$

The operation which combines the two functions f_X and f_Y in this fashion is called **convolution**. In other words, the PDF of the sum of two independent random variables is the convolution of their two PDFs.

Sum of Gaussian random variables

- Use the result derived earlier regarding the PDF of the sum of two independent random variables to show that the PDF of the sum of two independent Gaussian random variables is also Gaussian.
- Hint1: You have to take the convolution of two Gaussian PDFs. Try first with zero mean Gassian random variables.
- Hint2: Use the convolution theorem to see if the problem can be solved in the Fourier domain more easily; You will need to get the Fourier transform of a Gaussian PDF first.

Simple identities Pt. 1

- Show that the covariance of two statistically independent random variables is always zero
  - First, show that $E\big[X - E[X]\big] = 0$ (Hint: $E[X]$ is a constant)
  - Next, use the definition of covariance

$$C_{XY} = E[(X - E[X])(Y - E[Y])]$$

  and the definition of statistical independence (in terms of the joint pdf) along with the above result.